

ARNAB BANERJEE

MSc in Big Data Analytics, RKMVERI, Belur

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GitHub

Portfolio

LinkedIn



SUMMARY

MSc in Big Data Analytics student with a background in Mathematics and interests in distributed AI, machine learning, federated learning, speech AI, and LLM systems. Experienced with PyTorch, LangGraph, FastAPI, and Neo4j through research and applied AI projects.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)

MSc in Big Data Analytics, 2024–Present

Dakshin Dinajpur University

MSc in Mathematics, 2021–23

CGPA – 8.72

University of Calcutta

BSc in Mathematics, 2017–21

Score – 78.6%

SKILLS

- Languages:** Python, SQL, JavaScript, R, Bash, LaTeX
- AI/ML & Agentic Systems:** PyTorch, Transformers, NLP, LLMs, RAG, ASR, RL, Federated Learning, Gen-AI, LangChain, LangGraph, FastAPI, Neo4j, Vector Search, Prompt Engineering
- Data & Infrastructure:** NumPy, Pandas, Scikit-Learn, Spark, Kafka, Ray, Dask, Hadoop, PostgreSQL, SQLite, Git/GitHub, Linux, REST APIs, Jupyter Notebook, Playwright

COURSE WORK

- DSA, Linear Algebra & Matrix Computation, Statistics, Probability & Stochastic Processes, Machine Learning, Universal Human Values
- Java & Hadoop, Time Series & Survival Analysis, DL & NLP, Communicative English, Emerging Technologies
- Graph Databases & Distributed Computing, Multivariate Statistics, Database Management Systems, Introduction to Econometrics & Finance, Deep RL, Mining of Massive Datasets

LEADERSHIP & IMPACT

Co-founder, Clava Mathematica 2019 – Present

Built and helped grow a 500+ member mathematics community.

Event Lead, Riddle Run (Perceptron 2026)

Led the flagship treasure-hunt event and coordinated 10+ volunteers.

SUMMER INTERNSHIP

Summer Research Project under Dr. Indrila Ganguly

June 2026 – Present

Research Mentor: Dr. Indrila Ganguly (Fred Hutchinson Cancer Center)

- Conducting research on federated learning systems, uncertainty quantification, bootstrap-based statistical inference, and generalized linear models (GLMs).
- Exploring scalable statistical methodologies for distributed learning and inference under decentralized data settings.

PROJECTS

Causal Adaptive Streaming Decoder for Imagined Handwriting BCIs

RKMVERI

2026

Designed and implemented a causal GRU-based streaming decoder for imagined handwriting Brain–Computer Interfaces (BCIs), enabling adaptive real-time character prediction from neural signals. Achieved a 72.6% relative Character Error Rate improvement over a fixed-window decoding baseline while maintaining low decision latency.

Federated Clinical Copilot

RKMVERI

2026

Developing a privacy-preserving federated AI framework for clinical decision support across decentralized healthcare environments. Exploring distributed training methodologies, statistical inference, and uncertainty-aware healthcare AI systems.

Whisper-small-Benglish

Personal Project

2024

Fine-tuned OpenAI Whisper-small for Bengali-English code-switched speech recognition. Developed multilingual ASR workflows for transcription, evaluation, and deployment of Benglish speech systems.

Hindi-EmoTune

RKMVERI

2025

Developed a training-free framework for voice cloning and emotion-controlled Hindi speech synthesis. Modeled latent subspaces in XTTS for zero-shot emotional transfer and expressive speech generation.

ClanQuest

RKMVERI

2025

Built a multi-agent reinforcement learning simulation exploring emergent behaviors, emotional dynamics, resource competition, and clan interactions in a constrained geopolitical environment.

KaalDrishti AI

Personal Project

2025

Built an agentic AI Vedic astrology platform integrating Swiss Ephemeris, Neo4j knowledge graphs, vector retrieval, hybrid RAG pipelines, and LLM-based reasoning for personalized astrological analysis.